

Israel / hybrid

defense clearance

● open to the next challenge

18 years “on the metal” in low-level, real-time, product-grade engineering - from defense simulation and embedded Linux to firmware research, reverse engineering, board bring-up and hardware-accelerated edge AI of product-grade pipelines for robotic platforms.

Now one layer up: **AI-directed engineering**: I design the systems and the contracts, and direct swarms of agents that build to my standard - under one law: *contract before code*.

Target: a seat where agent leverage turns architectural ownership into real-world impact.

EXPERIENCE

Embedded AI Engineer Government of Israel · Robotics & Edge Division 2024 - now

- ▶ Edge-AI products for robotic platforms - board bring-up to full inference pipelines on embedded NPU / accelerator hardware , covering vision workloads and on-device LLM demos: Jetson, Hailo, Coral, Rockchip, Qualcomm, ModalAI, Stm, M5, Esp, Analog.
- ▶ Designed robotic processing-host and drone-swarm architectures - edge detection, comms, distributed algorithms, and AI-driven multi-agent development workflows.

R&D Engineer Government of Israel · Innovation Division 2022 - 2024

- ▶ Board bring-up of NPU-based computer-vision platform on i.MX 8M Plus - local CV inference directly on the integrated NPU , latter combined with hailo8.
- ▶ Created the platform model zoo, feasibility demos, and product-oriented proof points that validated edge inference for future embedded observation systems.

Embedded Linux Engineer Government of Israel · Firmware Division 2018 - 2022

- ▶ Developed Embedded-Linux / BSP infrastructure for new products, leading board bring-up across i.MX, STM32, Qualcomm, and custom platforms - boot flow, kernel, drivers, firmware flashing, HW/SW integration.
- ▶ Integrated RF, SDR, wireless modules, cellular modems, sensors, and storage components; supported low-level Wi-Fi, Bluetooth, RF, and cellular-stack firmware research, and drove full system validation.

Security Researcher & Team Lead Sigmabit · Cyber Division 2016 - 2018

- ▶ Built and led the embedded-security research domain - Linux device reverse , firmware analysis, kernel debugging, protocol research, wireless equipment.
- ▶ Researched vulnerabilities and built controlled proof-of-concept exploits - remote access, malfunction, misuse, and kernel-level behavior; network-booted custom kernels into closed-filesystem targets.

Embedded SW Engineer Elbit / Elisra · Airborne EW 2014 - 2016

- ▶ Real-time C++ VxWorks operational software for a miniaturized airborne EW system on CPU+FPGA architecture - BSP/firmware layers, RF front , VxWorks user/kernel-space.
- ▶ Initiated Zynq SoC adoption to drive system miniaturization; joined the cross-company hardware bring-up excellence team.

Embedded SW Developer IAI Elta · Trophy Active Protection System 2011 - 2014

- ▶ Real-time embedded C / C++ on a defense-grade Linux platform for the Trophy APS. System analysis and validation of mission-critical cross-system operational.
- ▶ Designed the black-box recorder that improved tracking-algorithm analysis ,captured operational logic and runtime behavior for system-performance proof.

C++ Developer MoD / IDF · Survivability Branch 2008 - 2011

- ▶ Linux / C++ 3D simulation systems and CUDA-accelerated distributed infrastructure for survivability research, operational analysis, and armor attack/defense optimization
- ▶ Extended armor framework into an innovation-award-winning line-of-sight situational-awareness capability while contributed to ballistics research with HPC workflow.

INTERESTS

Maker across materials and methods: pushing hot plastic through a nozzle in 3D, building with wood, metal, glass, welding, cutting, reshaping, blacksmithing, and reusing materials into practical workspaces and tools. | **Mentor** young builders through LEGO robotics / FLL

“molding young minds brick by brick” while teaching think, build, test, and improve. | **Swim, Code, Repeat.**

LOOKING FOR

- ▶ A hard technical mission - autonomy, defense, robotics, edge AI, embedded.
- ▶ Ownership through architecture - broad domain responsibility.
- ▶ AI-native engineering in production.

EDUCATION & TRAINING

- ▶ 2025 · Software Architecture · DevExpress
- ▶ 2024 · AI Developer Program · Technion
- ▶ 2021 · Ghidra Reverse Engineering · hardware.io
- ▶ 2020 · Hardware Hacking · ChipWhisperer · Hands-On Training
- ▶ 2019 · Cellular Networks 2G→5G · Logtel
- ▶ 2019 · Embedded Systems Security Bootcamp · Hands-On Training
- ▶ 2019 · ARM Embedded-Systems Security · Hands-On Training
- ▶ 2018 · IoT Firmware & RTOS Exploitation · TNS
- ▶ 2017 · Developing Linux Device Drivers · Linux Foundation
- ▶ 2016 · Linux Kernel Internals & Debugging · Linux Foundation
- ▶ 2015 · VxWorks Essentials · Wind River
- ▶ 2015 · OpenCL for Altera SoCs
- ▶ 2015 · Zynq SoC Architecture · Logtel
- ▶ 2014 · Real-Time C++ Application
- ▶ 2014 · Real-Time C++ Design Patterns
- ▶ 2013 · Embedded Linux Design · RTED
- ▶ 2011 · Linux Essentials · Technion
- ▶ 2008 · B.Sc. Mechanical Engineering, Robotics minor · Ben-Gurion

CORE STACK

on the metal - C / C++ | Embedded Linux | kernel | device drivers | Yocto | BSP | VxWorks | RTOS | Zynq | FPGA | i.MX | STM32 | Qualcomm | NPU | edge inference | Jetson | Hailo | Rockchip | ST | quantization | distributed systems | reverse engineering | SDR / RF |

ai-directed engineering - systems architecture for agent swarms, plan-driven development, multi-agent runners, data contracts & quality gates, closed learning loops, local-first tooling.